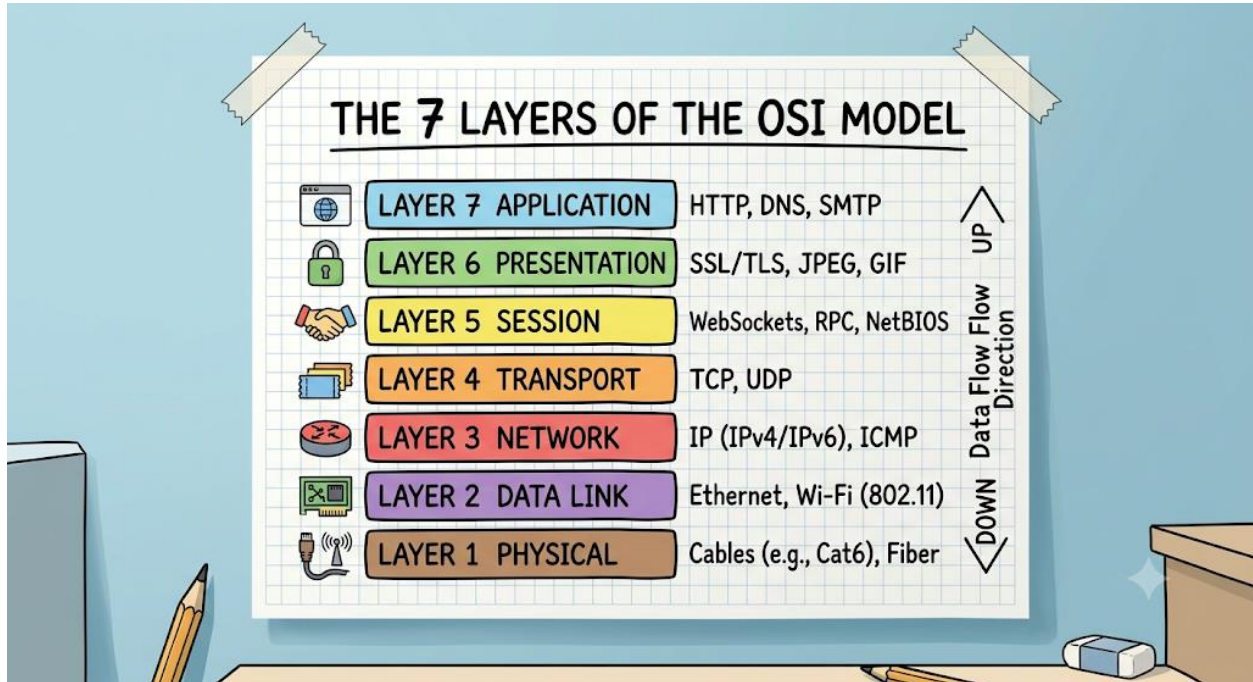


ASSIGNMENT : 11 – OSI AND NETWORK PROTOCOLS

GUIDED BY : VIPUL MISTRI

Q.1 Draw the OSI model on paper or using any drawing tool, label all 7 layers, and write one real-world example protocol (like HTTP, TCP, IP, etc.) next to each layer.

ANS :



Q.2 Create a table listing 5 popular apps you use (such as WhatsApp, Instagram, Zomato, Flipkart, Paytm) and for each, identify two network protocols that are likely used when you interact with the app (for example, HTTPS, TCP, UDP, DNS, etc.).

ANS :

Application	Protocol 1	Protocol 2
WhatsApp	WebSockets / TLS (for real-time messaging and encrypted data transport)	SRTP / UDP (for low-latency voice and video calls)
Instagram	HTTPS (HTTP/2 or HTTP/3) (for fetching feeds, images, and user data securely)	DNS (to resolve api.instagram.com to an IP address)

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Application	Protocol 1	Protocol 2
Zomato	HTTPS (for browsing menus, authenticating, and placing orders)	TCP (to ensure reliable delivery of order and payment data)
Flipkart	HTTPS (for encrypted transactions, product catalog loading, and search)	TCP (to prevent packet loss during checkout and payment gateway processing)
Paytm	HTTPS / TLS (for secure bank-level encryption during UPI and wallet payments)	DNS (for fast server endpoint resolution during quick QR code scans)

Q.3 Use the command prompt or terminal to run 'ping' and 'tracert' (Windows) or 'traceroute' (Linux/Mac) to www.youtube.com, then explain which OSI layers are involved in these commands and how.

ANS : Here is how to run the commands and how they interact with the OSI model:

Step 1: Commands to Run

Open your **Command Prompt** (Windows) or **Terminal** (Linux/macOS) and run:

On Windows:

```
ping www.youtube.com
```

```
tracert www.youtube.com
```

Bash

```
ping -c 4 www.youtube.com
```

```
traceroute www.youtube.com
```

Step 2: OSI Layers Involved in ping and tracert / traceroute

Both tools operate across multiple OSI layers to test network reachability and path diagnostics:

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1. Layer 7: Application Layer (DNS Trigger)

- **How it's involved:** Although ping and tracert operate as system utilities, when you pass a domain name like www.youtube.com, the system invokes **DNS (Domain Name System)** to resolve the domain into an IPv4 or IPv6 address before sending packets.

2. Layer 3: Network Layer (Core Operation — ICMP & IP)

- ping: Uses **ICMP (Internet Control Message Protocol) Echo Request** (Type 8) and **Echo Reply** (Type 0) messages wrapped directly inside **IP packets**. ICMP operates at Layer 3 to verify host availability and measure Round-Trip Time (RTT).
- tracert / traceroute: Manipulates the **TTL (Time to Live)** field in the IP header, starting from TTL=1 and incrementing by 1 for each hop. When a router receives a packet with TTL=1, it drops the packet and responds with an **ICMP Time Exceeded** (Type 11) message, allowing tracert to discover each hop along the path to YouTube's server.

3. Layer 4: Transport Layer (Linux/macOS traceroute & TCP/UDP)

- **Windows** tracert: Uses raw **ICMP Echo Requests** at Layer 3.

4. Layer 2: Data Link Layer (Ethernet / Wi-Fi)

- Encapsulates the IP packets into **Ethernet frames** or **802.11 Wi-Fi frames** using **MAC (Media Access Control)** addresses to move packets hop-by-hop between your computer and default gateway/routers.

5. Layer 1: Physical Layer (Hardware/Medium)

- Converts frames into physical signals (electrical signals over copper Ethernet cabling, light pulses over optical fiber, or radio waves over Wi-Fi) for transmission.

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Q.4 Write a short scenario describing how a message sent from your device to a friend's device on WhatsApp travels through each OSI layer, mentioning what happens at each step. Hint: Start from Application layer (your message in WhatsApp) down to Physical layer (actual data sent over WiFi/mobile network), then back up to Application layer on your friend's phone.

ANS : Here is how a WhatsApp message ("Hey!") travels from your phone to your friend's phone through all 7 OSI layers:

Phase 1: Descending the OSI Model (Sender's Device)

1. Layer 7: Application Layer

- **Action:** You type "Hey!" in WhatsApp and tap **Send**. The app formats the message payload into an internal JSON/protocol data unit and triggers the transmission request.

2. Layer 6: Presentation Layer

- **Action:** WhatsApp performs **End-to-End Encryption (E2EE)** using the Signal Protocol, converting your plain text "Hey!" into unreadable ciphertext. It also compresses the payload for efficient network delivery.

3. Layer 5: Session Layer

- **Action:** Manages and maintains the active, persistent **WebSocket connection** session between your WhatsApp client and WhatsApp's primary edge servers.

4. Layer 4: Transport Layer

- **Action:** Breaks the encrypted data into segments and attaches a **TCP header** specifying the source and destination port numbers (e.g., HTTPS/WebSocket port 443) to guarantee reliable, ordered delivery.

5. Layer 3: Network Layer

- **Action:** Encapsulates the TCP segment into an **IP packet**, adding your device's source IP address and the destination server's IP address.

6. Layer 2: Data Link Layer

- **Action:** Wraps the IP packet into an **Ethernet or 802.11 Wi-Fi frame**, attaching your device's source MAC address and your Wi-Fi router's destination MAC address for local physical node hop.

7. Layer 1: Physical Layer

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- **Action:** Converts the digital binary bits (0s and 1s) into physical **radio signals** (over Wi-Fi/4G/5G) or electrical/light signals (fiber cables) to send data across the internet to WhatsApp's server infrastructure.

- **Phase 2: Transiting the Server**

- WhatsApp's server receives the encrypted packet, processes the recipient's connection state, and forwards the encrypted payload back down its own network stack toward your friend's device IP.

Phase 3: Ascending the OSI Model (Friend's Device)

1. Layer 1: Physical Layer

- **Action:** Your friend's phone antenna receives the incoming radio signals and converts them back into raw binary bits.

2. Layer 2: Data Link Layer

- **Action:** Strips off the MAC header, verifies frame integrity via CRC checks, and extracts the IP packet.

3. Layer 3: Network Layer

- **Action:** Inspects the IP header to confirm their device IP matches the destination IP, then strips off the IP header.

4. Layer 4: Transport Layer

- **Action:** Checks TCP sequence numbers and port numbers to ensure no missing data, then passes the assembled segment up to the application listening on port 443.

5. Layer 5: Session Layer

- **Action:** Identifies the active WhatsApp session and passes the encrypted data stream to the client.

6. Layer 6: Presentation Layer

- **Action:** Uses your friend's private key to **decrypt** the ciphertext back into plain text and decompresses the payload.

7. Layer 7: Application Layer

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- **Action:** WhatsApp renders "Hey!" on your friend's screen and plays the incoming message notification chime.